

EXAM 7 – SPRING 2012

10. (2.75 points)

Given the following information for two proportional casualty treaties as of December 31, 2011:

<u>Treaty ID</u>	<u>Inception Date</u>	<u>Treaty Coverage Basis</u>	<u>Written Premium</u>	<u>Aggregate Reported Loss</u>
A	Jan 1, 2009	Loss Occurring	\$ 80,000	\$ 50,000
B	Jul 1, 2009	Policies Written	100,000	20,000

<u>Age of Accident Year (in months)</u>	<u>Age to Ultimate LDF</u>
36	1.25
24	2.00
12	3.00

- Both treaties have a one-year term.
- All underlying policies are one-year, occurrence-based policies with effective dates uniformly distributed over the year.
- Treaty A commission, brokerage and internal expense is 5% of premium.
- Treaty B commission, brokerage and internal expense is 20% of premium.
- The loss development factors apply to all exposures in the applicable accident year.

a. (2 points)

Estimate the ultimate losses for the two treaties combined using the Stanard-Bühlmann method.

b. (0.25 point)

Identify a situation in which the chain ladder method is preferred over the Stanard-Bühlmann method when estimating IBNR for a reinsurer with long-tail exposures.

c. (0.25 point)

Identify a situation in which the Stanard-Bühlmann method is preferred over the chain ladder method when estimating IBNR for a reinsurer with long-tail exposures.

d. (0.25 point)

Identify the key innovation of the Stanard-Bühlmann method over the Bornhuetter-Ferguson method.

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Question 10 Sample Answer

Solution 1

- a) Remove commission, brokerage, & internal expenses

$$A \ 80,000 \times (1-.05) = 76,000$$

$$B \ 100,000 \times (1-.2) = 80,000$$

At 12/31/11, treaty A is 36 months mature

treaty B is 24 months mature (avg written date = 1/1/10)

$$\begin{aligned} \text{So used-up premium} &= 76,000(1/1.25) + 80,000 (1/2) \\ &= 100,800 \end{aligned}$$

$$\text{Total losses} = 70,000 \quad \text{so ELR} = 70/100.8 = .694$$

$$\text{IBNR} = .694 (76(1- (1/1.25) + 80(1- 1/2))) = 38,333.33$$

$$\text{Ult loss} = \text{IBNR} + \text{rep to date} = \$108,333.33$$

- b) CL is preferred when it is difficult to bring premium from all yrs to same rate level.
- c) SB is preferred when there are fluctuations in rep losses but we expect ELR (to on-level earned premium) has remained constant.
- d) ELR is based on loss data, not just judgmentally selected.

Solution 2

- a)

		@12/31/11 % Earned	Adj. Prem	
A	loss occurring (1/1/09–12/31/09)	100%	80(.95)=76K	@36 mo
B	pol written (7/1/09–6/30/09)	100%	80K	1/8@36 3/4@24 1/8@12

	Prem		% Rpt	Used Prem
2009	76 + (1/8)(180k)	= 86,000	1/1.25 = .8	68,800
2010	(3/4)(180k)	= 60,000	.5	30,000
2011	(1/8)(180k)	= 10,000	.333	3,000
		= 156,000		102,133

$$\text{ELR} = (50k + 20k) / 102,133 = .685$$

$$(.685)(156,000) = 106,910$$

- b) When the premium is difficult to adjust to current rate level or net the expenses, it is more appropriate to use chain ladder.
- c) When the Stanard-Bühlmann method is preferred to the chain ladder.
- d) Stanard-Bühlmann uses actual experience to estimate the expected loss ratio, where the Bornhuetter Ferguson method uses an apriori loss ratio.

Examiner Comment

Many candidates lost points for not removing expenses from the premium. Almost all candidates lost points for not correctly allocated the treaty premiums and losses over the 3 year subject period (AY's 09 – 11). If the development of the SB ELR and Ultimate Losses followed appropriate form, even without expense removal or proper AY allocation, full credit was available for that step. If there was no distinction in lag factors by AY, partial credit was subtracted. No credit was given for candidates that did not know to add the 2 contracts together for determining IBNR.

For b., c., and d., candidates answered fairly accurately. However, a number of candidates didn't relate the need for premiums at current rate levels as the main disadvantage of the SB method.